

Elastic Stack (Elasticsearch, Logstash, Kibana, Filebeat) Installation Guide

This guide explains how to install and configure Elastic Stack 8.x on Ubuntu.

1. Update and Upgrade System Packages

```
sudo apt update \&\& sudo apt upgrade -y
```

2. Install Java (Required for Elasticsearch and Logstash)

```
sudo apt install openjdk-17-jdk -y
```

Check java version

```
java -version
```

Set up ELK artifacts

```
sudo apt install apt-transport-https ca-certificates curl gnupg -y  
curl -fsSL [https://artifacts.elastic.co/GPG-KEY-elasticsearch]  
(https://artifacts.elastic.co/GPG-KEY-elasticsearch) | sudo gpg --  
dearmor -o /usr/share/keyrings/elastic-archive-keyring.gpg  
echo "deb [signed-by=/usr/share/keyrings/elastic-archive-  
keyring.gpg] [https://artifacts.elastic.co/packages/8.x/apt]  
(https://artifacts.elastic.co/packages/8.x/apt) stable main" |  
sudo tee /etc/apt/sources.list.d/elastic-8.x.list
```

Install Vim editor

```
sudo apt install vim -y
```

Install Elasticsearch

```
sudo apt update  
sudo apt install elasticsearch -y
```

Edit elasticsearch configuration to disable security

```
sudo vi /etc/elasticsearch/elasticsearch.yml
```

Update these settings to false, to disable security

```
xpack.security.enabled: false  
xpack.security.transport.ssl.enabled: false  
xpack.security.http.ssl.enabled: false  
network.host: 0.0.0.0
```

Set up systemctl for elasticsearch

```
sudo systemctl daemon-reload  
sudo systemctl enable elasticsearch  
sudo systemctl start elasticsearch
```

Check elastic search is running now

```
curl http://localhost:9200
```

You should be able to see some output like this

```
{  
  "name" : "Edhitas-MacBook-Air.local",  
  "cluster_name" : "elasticsearch_roopal",  
  "cluster_uuid" : "XskTKQYUQKKdMZSDVky7fg",  
  "version" : {  
    "number" : "7.17.4",  
    "build_flavor" : "default",
```

```
"build_type" : "tar",
"build_hash" : "79878662c54c886ae89206c685d9f1051a9d6411",
"build_date" : "2022-05-18T18:04:20.964345128Z",
"build_snapshot" : false,
"lucene_version" : "8.11.1",
"minimum_wire_compatibility_version" : "6.8.0",
"minimum_index_compatibility_version" : "6.0.0-beta1"
},
>tagline" : "You Know, for Search"
}
```

Install logstash

```
sudo apt install logstash -y
```

Setup permissions

```
chmod -R 777 /usr/share/logstash/data
```

Install filebeat

```
sudo apt install filebeat -y
```

Setup permissions

```
chmod -R 777 /usr/share/filebeat/data
```

Install Kibana

```
sudo apt install kibana -y
```

Remove security from kibana

```
sudo vi /etc/kibana/kibana.yml
```

```
server.host: "0.0.0.0"
xpack.security.enabled: false

sudo systemctl enable kibana
sudo systemctl start kibana
```

Install git

```
sudo apt install git
```

Clone the repo

```
git clone [https://github.com/mirajgodha/elastic-stack.git]
(https://github.com/mirajgodha/elastic-stack.git)
```

Installation Complete

Elasticsearch should be accessible at <http://localhost:9200>

Kibana should be accessible at <http://localhost:5601>